NEXT STEPS: ROADMAP TO EVIDENCE-AWARE ECG AI

The Three Pillars: Clinical Knowledge Integration, Deterministic Ternary Grounding Gate, and Controlled Comparison Protocol

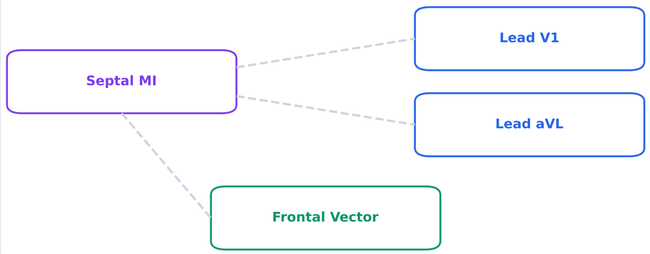
SPOKEN SCRIPT: "***Our next steps are to develop the clinical knowledge-assisted path, improve the evidence-grounding checks, and conduct a controlled comparison under a properly defined evaluation protocol.***"

PILLAR 1: KNOWLEDGE PATH

Electrophysiological Graph Priors

Structured SNOMED-CT Subgraphs

Curated clinical anatomical relations



CORE TECHNICAL MECHANISM:
- Integrates expert clinical relations into the cross-attention reasoning layer.
- Re-routes attention: When primary chest leads are masked, priors guide the model to examine observable limb lead deviations (e.g., reciprocal depression in aVL).
- Validated Impact: Delays Reasoning Failure Boundary down to 2-lead telemetry (+15.7% AURDC).

PILLAR 2: GROUNDING CHECKS

Deterministic Ternary Verification Gate

Ternary Evidence Classification

Auditing every clinical claim against active sensors

[SUPPORTED]
Sensor Active & Evidence Verified

[CONTRADICTED]
Active Lead Disproves Finding

[UNAVAILABLE]
Required Sensor Physically Absent

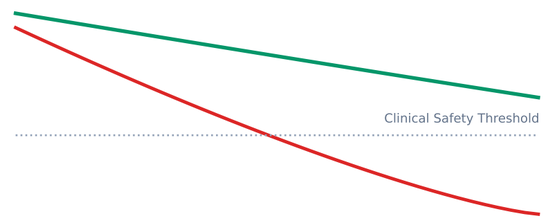
CORE TECHNICAL MECHANISM:
- Cross-checks each reasoning token against the deterministic hardware mask vector m in {0,1}^12.
- Zero-Tolerance for Hallucinations: If top cited leads are marked UNAVAILABLE, prediction is flagged as ungrounded before clinician review.
- Provides mathematical True Grounding Ratio to replace deceptive Macro-AUROC metrics.

PILLAR 3: CONTROLLED PROTOCOL

Standardized Clinical Benchmark Suite

Evaluation Metrics & Safety Boundaries

Multi-cohort stress-testing (36,957 records)



CORE TECHNICAL MECHANISM:
- Rigorous Head-to-Head Protocol: Compare Control vs. Knowledge-Assisted models across 7 lead levels.
- Formal Safety Metrics: Formally isolates the Reasoning Failure Boundary (RFB) and Area Under the Reasoning Degradation Curve (AURDC).
- Pre-Hospital Validation: Evaluates readiness for real-world ambulance telemetry & wearables.

TARGET OUTCOME: An Auditable, Evidence-Aware Standard for Medical AI Evaluation